

Question description

There are three common ways to traverse the nodes of a binary tree:

- *Preorder*: First process the root, then the left subtree, and finally the right subtree.
- *Inorder*: First process the left subtree, then the root, and finally the right subtree.
- *Postorder*: First process the left subtree, then the right subtree, and finally the root.

There is a binary tree of n nodes with distinct labels. You are given the preorder and inorder traversals of the tree, and your task is to determine its postorder traversal.

Constraints

- $1 \leq n \leq 10^5$

Input

The first input line has an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

After this, there are two lines describing the preorder and inorder traversals of the tree. Both lines consist of n integers.

You can assume that the input corresponds to a binary tree.

Output

Print the postorder traversal of the tree.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
5 3 2 1 4
3 5 1 2 4

EXPECTED OUTPUT

3 1 4 2 5

Test Case 2

INPUT (STDIN)

6
6 5 3 2 1 4
6 3 5 1 2 4

EXPECTED OUTPUT

3 1 4 2 5 6

▼ Mandatory Test Cases

Test Case 1

KEYWORD

for(i=1;i<=n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

390

Test Case 3

NLOC

41